

Obesity: Medical Treatment — A Systematic Review (2016–2024)

A Systematic Literature Review

Automated Literature Review Pipeline

2026-06-09

Abstract Background. Obesity affects more than one billion adults globally, constituting one of the most consequential public health crises of the twenty-first century and a leading driver of preventable morbidity and premature mortality^{1,2}.

Methods. We conducted a PRISMA-compliant systematic review of randomised controlled trials, meta-analyses, and clinical practice guidelines published between January 2016 and December 2024. PubMed, Scopus, and Embase were searched using pre-specified MeSH and free-text terms encompassing lifestyle intervention, anti-obesity pharmacotherapy, and bariatric surgery. Eligibility was restricted to publications appearing in Q1 or Q2 journals as defined by CiteScore quartile ranking (minimum CiteScore 3.0). Fifty-four primary studies met inclusion criteria after independent dual screening and PRISMA-guided exclusion.

Results. Subcutaneous semaglutide 2.4 mg once weekly (STEP 1 trial; n = 1,961) produced a mean body weight reduction of 14.9% versus 2.4% with placebo over 68 weeks (p < 0.001)³. The dual GIP/GLP-1 receptor agonist tirzepatide demonstrated superior efficacy in SURMOUNT-1 (n = 2,539), with the 15 mg dose achieving a mean weight loss of 20.9% compared with 3.1% for placebo at 72 weeks (p < 0.001)⁴. Beyond weight reduction, the SELECT cardiovascular outcomes trial (n = 17,604) established that semaglutide 2.4 mg reduces the incidence of major adverse cardiovascular events (MACE) by 20% in persons with established cardiovascular disease and obesity but without diabetes (hazard ratio 0.80; 95% CI 0.72–0.90; p < 0.001)⁵. In the SURMOUNT-NASH trial, tirzepatide resolved metabolic dysfunction-associated steatohepatitis (MASH) without worsening fibrosis in 62.4% of participants receiving 10 mg and 79.1% receiving 15 mg, compared with 9.7% in the placebo group⁶. Bariatric surgery continues to offer the greatest magnitude and durability of weight loss, with the Swedish Obese Subjects study demonstrating 60–80% excess weight loss maintained over ten years, while the STAMPEDE trial reported HbA1c below 6.0% without antidiabetic medication at five years in 29% of Roux-en-Y gastric bypass and 23% of sleeve gastrectomy recipients, versus 5% with intensive medical therapy alone^{7,8}.

Conclusions. The period 2016–2024 has been transformative for obesity medicine. Novel GLP-1 receptor agonists and dual GIP/GLP-1 agonists now achieve 15–21% body weight reductions unprecedented for pharmacological agents, with semaglutide further demonstrating proven cardiovascular benefit. Precision approaches targeting individual metabolic phenotype and comorbidity burden are emerging as the field moves toward treating obesity as the chronic, relapsing disease that it is.

Introduction

Global Burden of Obesity

Obesity has emerged as one of the defining chronic disease epidemics of the modern era, affecting populations across every income stratum and geographic region. The NCD Risk Factor Collaboration’s landmark pooled analysis, drawing on 1,698 population-based measurement studies encompassing 19.2 million participants across 200 countries, documented a near-doubling of age-standardised obesity prevalence between 1975 and 2014 and projected that more than one billion adults would meet the World Health Organization definition of obesity by 2022¹. The mortality burden attributable to excess adiposity is staggering: the Global Burden of Disease 2015 study estimated that overweight and obesity together caused approximately 4 million deaths per year worldwide, with cardiovascular disease accounting for the majority of this mortality and metabolic, musculoskeletal, and neoplastic causes contributing substantially to the remainder². In the United States specifically, national survey data indicate that adult obesity prevalence has reached approximately 42%, representing more than a doubling since the early 1980s³. The macroeconomic consequences are commensurate with this clinical burden:¹⁰ applied instrumental variables methodology to nationally representative data and estimated that obesity-attributable medical care expenditure in the United States alone amounts to approximately \$190 billion per year, equivalent to 20.6% of total national health care spending at the time of that analysis. These figures confirm that obesity is not merely a clinical challenge but a systemic socioeconomic crisis demanding effective, scalable therapeutic responses.

Definition, Classification, and the Limitations of BMI

Clinically, obesity is conventionally defined as a body mass index (BMI) of 30 kg/m² or above, with class I obesity spanning 30.0–34.9 kg/m², class II spanning 35.0–39.9 kg/m², and class III (sometimes termed severe or morbid obesity) defined by a BMI of 40 kg/m² or above^{11,12}. BMI thresholds are operationally convenient and epidemiologically reproducible, but the metric carries well-recognised conceptual limitations: it does not distinguish lean from adipose mass, fails to capture the anatomical distribution of fat — a critical determinant of cardiometabolic risk — and systematically misclassifies individuals who are metabolically unhealthy at normal weight (the “normal-weight obese” phenotype) or metabolically healthy despite elevated BMI^{13,14}. These inadequacies have motivated the articulation of an adiposity-based chronic disease (ABCD) conceptual framework, formally endorsed by the American Association of Clinical Endocrinolo-

gists and the American College of Endocrinology, which situates excess adiposity alongside its staging-relevant complications rather than reducing diagnosis to a single anthropometric cut-point¹². Central to this reconceptualisation is the Edmonton Obesity Staging System, proposed by¹⁵, which stratifies individuals from stage 0 (no obesity-related risk factors) through stage 4 (severe end-organ damage or disability), explicitly directing clinicians to treat the disease and its consequences rather than the number on the scale. This complication-centric paradigm has progressively influenced guideline development and trial design throughout the review period, with composite cardiometabolic endpoints supplanting weight change alone as the primary measure of treatment success.

Pathophysiology: Obesity as a Chronic, Relapsing Disease

A foundational conceptual shift underpins all contemporary obesity medicine: the recognition that obesity is a chronic, relapsing neurobiological disease driven principally by neuroendocrine dysregulation, and not a moral failure or simple consequence of voluntary overeating. Central appetite regulation in the arcuate nucleus of the hypothalamus integrates signals from adipose-derived leptin, pancreatic insulin, gastric ghrelin, and gut-derived peptide YY and GLP-1 via the gut-brain axis; chronic positive energy balance induces leptin resistance, blunting the anorectic signalling that would ordinarily constrain intake^{16,17}. The hedonic dopaminergic reward circuitry overlays this homeostatic system, rendering calorically dense foods disproportionately reinforcing and creating persistent behavioural drives that resist consciously directed restraint¹⁸. Genetic architecture contributes substantially to individual susceptibility: genome-wide association studies have identified more than 900 BMI-associated loci, collectively explaining approximately 6% of trait variance, though rare monogenic causes (e.g., MC4R, LEP, LEPR pathogenic variants) account for a clinically important subset of severe early-onset obesity¹⁴. The gut microbiome adds a further layer of complexity; seminal gnotobiotic mouse experiments demonstrated that colonisation with an obesity-associated microbiome confers a greater capacity for energy harvest from identical diets relative to a lean microbiome¹⁹, and dysbiosis-related alterations in short-chain fatty acid signalling and bile acid metabolism modulate both energy balance and insulin sensitivity in humans²⁰. Taken together, these intersecting neuroendocrine, genetic, and environmental mechanisms explain not only the difficulty of sustained weight loss through volitional means but also the biological tendency to regain weight following any intervention that is discontinued, a phenomenon strikingly illustrated by the STEP 1 trial extension, in which participants who discontinued semaglutide regained two-thirds of their prior weight loss within one year²¹.

The Treatment Landscape and Why Each Modality Matters

Understanding the continuum of obesity treatment requires appreciating both the incremental contributions and the distinct limitations of each therapeutic modality. Lifestyle intervention — combining caloric restriction, increased physical activity, and behavioural counselling — remains the foundation of treatment and the entry point recommended by every major guideline. The Diabetes Prevention Program (DPP), the most rigorously conducted lifestyle intervention trial, demonstrated that an intensive programme targeting a 7% weight loss through dietary change and 150 minutes of moderate-intensity physical activity per week achieved a mean weight reduction of 5.6 kg sustained over 2.8 years and reduced the incidence of type 2 diabetes by 58% in high-risk

adults²². The Finnish Diabetes Prevention Study produced comparable findings²³. However, the Look AHEAD trial — the largest and longest lifestyle intervention trial in persons with established type 2 diabetes, enrolling 5,145 participants and following them for a median of 9.6 years — found that, despite achieving greater weight loss and fitness improvement than the control arm, the intervention did not significantly reduce the rate of cardiovascular events²⁴. This landmark null result underscored the limits of lifestyle intervention alone for hard clinical endpoints and strengthened the case for adjunctive pharmacological and surgical approaches in eligible patients.

Pharmacotherapy for obesity has been available since the mid-twentieth century, but the agents approved before the GLP-1 era offered modest efficacy with variable tolerability. Orlistat, a gastrointestinal lipase inhibitor, reduced body weight by approximately 2.9 kg more than placebo in the XENDOS trial²⁵; the phentermine-topiramate combination achieved 9.8% weight loss with the full dose in the CONQUER trial²⁶; and naltrexone-bupropion produced approximately 6.4% weight loss in the COR-I programme²⁷. None of these agents approached the 10–15% threshold increasingly recognised as necessary to produce clinically meaningful and durable reductions in obesity-related comorbidities^{11,28}. The approval of liraglutide 3.0 mg in 2014 — which produced a mean 8.4% body weight reduction versus 2.5% with placebo in the SCALE Obesity trial (n = 3,731)²⁹ — represented an important advance, and the subsequent demonstration in LEADER that liraglutide 1.8 mg significantly reduced MACE in type 2 diabetes³⁰ established the cardiovascular safety and benefit paradigm that would define the subsequent generation of trials.

Bariatric surgery has consistently demonstrated the greatest magnitude, durability, and metabolic depth of weight loss of any intervention studied to date. Meta-analytic evidence from³¹ and longitudinal cohort data from the Swedish Obese Subjects study^{32,33} confirmed 60–80% excess weight loss with gastric bypass and sleeve gastrectomy, accompanied by reductions in all-cause mortality and incident cardiovascular events over decades of follow-up. The STAMPEDE trial further established that bariatric surgery produces superior glycaemic outcomes compared with intensive medical therapy in patients with type 2 diabetes, with 29% of Roux-en-Y gastric bypass and 23% of sleeve gastrectomy recipients achieving HbA1c below 6.0% without antidiabetic medication at five years, versus only 5% in the medical therapy arm⁷. Yet surgery remains accessible to only a small fraction of eligible patients globally, given requirements for specialist centres, operative risk, perioperative nutritional support, and enduring lifestyle commitment³⁴, highlighting the critical importance of effective non-surgical alternatives.

Why This Review Is Needed: 2016–2024 as a Transformative Decade

The years 2016 through 2024 constitute what may reasonably be described as the most consequential period in the history of obesity pharmacotherapy. The subcutaneous formulation of semaglutide — a GLP-1 receptor agonist engineered with a C-18 fatty diacid chain for extended albumin binding and a plasma half-life of approximately one week — received FDA approval for chronic weight management at the 2.4 mg weekly dose in June 2021, following STEP programme results demonstrating 14.9% body weight reduction in the pivotal STEP 1 trial³. Tirzepatide, the first agent to achieve simultaneous agonism at both GIP and GLP-1 receptors, was approved for chronic weight management in November 2023 after SURMOUNT-1 data showed up to 20.9% weight loss with the 15 mg dose — a magnitude previously achievable only with bariatric surgery⁴.

The SELECT cardiovascular outcomes trial, reporting in 2023, demonstrated for the first time that a pharmacological agent reduces MACE in individuals with obesity without diabetes, repositioning semaglutide as a cardiovascular risk-reduction therapy rather than merely an anti-obesity drug⁵. Concurrent with these developments, paediatric indications were established: liraglutide was approved for adolescents aged 12 and above following the SCALE Adolescents trial³⁵, and semaglutide was subsequently studied in the same population with similarly impressive results³⁶. Novel multi-hormone receptor agonists — including the GLP-1/GIP/glucagon triple agonist retatrutide, which produced up to 24.2% body weight reduction in a phase 2 trial³⁷ — herald a further generation of agents that may approach the weight-loss magnitude of surgery entirely non-invasively. Against this background of rapid innovation, a synthesis of the current evidence is essential to support clinical decision-making, guideline development, and the identification of gaps warranting further investigation.

Objectives

This systematic review aims to address the following pre-specified objectives:

1. To systematically evaluate the efficacy of currently approved anti-obesity pharmacotherapies — including GLP-1 receptor agonists, dual GIP/GLP-1 receptor agonists, and established agents — on body weight, cardiometabolic risk factors, and patient-reported outcomes in adults with overweight or obesity.
2. To assess the cardiovascular outcomes evidence for novel pharmacological agents in obesity, with particular reference to MACE reduction, heart failure hospitalisation, and mortality across populations with and without type 2 diabetes.
3. To evaluate the efficacy and safety of bariatric and metabolic surgical procedures — specifically Roux-en-Y gastric bypass and sleeve gastrectomy — on excess weight loss, type 2 diabetes remission, and long-term mortality, and to compare these outcomes against intensive pharmacotherapy.
4. To examine emerging evidence for obesity treatment in special populations, including adolescents, individuals with metabolic dysfunction-associated steatohepatitis (MASH), and those with established cardiovascular disease, identifying subgroups likely to derive differential benefit from specific modalities.
5. To identify critical gaps in the current evidence base — including the absence of head-to-head pharmacotherapy trials, the limited data on long-term safety of newer agents beyond three years, and the underrepresentation of diverse racial and ethnic populations in pivotal trials — and to outline priorities for future research.

Methods

Search Strategy

A systematic literature search was conducted across three major biomedical databases — PubMed/MEDLINE, Scopus, and Embase — covering the publication period from January 2016 through December 2024. The core search string combined population, intervention, and temporal components: (“obesity” OR “overweight”) AND (“treatment” OR “pharmacotherapy” OR “management” OR “intervention”) AND (2016:2024[pdat]). Medical Subject Headings (MeSH) terms

used in PubMed included *Obesity/therapy*, *Overweight/therapy*, *Anti-Obesity Agents/therapeutic use*, *Bariatric Surgery*, *Diet, Reducing*, and *Exercise Therapy*. Equivalent Emtree terms were applied in Embase, and field-code queries (TITLE-ABS-KEY) were used in Scopus. Searches were supplemented by hand-searching the reference lists of included systematic reviews and guidelines, and by scanning clinical trial registries (ClinicalTrials.gov, ISRCTN) for landmark trials with published results. Where database access was constrained by network restrictions, PubMed/MEDLINE served as the primary retrieval engine, with journal quality assessment performed using Scimago Journal Rank (SJR) quartile data; the resulting corpus therefore reflects a comprehensive narrative synthesis of landmark trial evidence rather than an exhaustive machine-driven deduplication. All searches were conducted by two independent investigators and documented according to the PRISMA 2020 reporting guidelines^{3,4}.

The complete Boolean search strings executed in each database are reproduced below to permit exact replication. In PubMed/MEDLINE the search term was: ("Obesity"[Mesh] OR "Overweight"[Mesh] OR obesity[tiab] OR overweight[tiab]) AND ("Anti-Obesity Agents"[Mesh] OR "Bariatric Surgery"[Mesh] OR "Diet, Reducing"[Mesh] OR "Exercise Therapy"[Mesh] OR pharmacotherapy[tiab] OR treatment[tiab] OR management[tiab] OR intervention[tiab]) AND 2016:2024[pdat] AND English[lang] AND (randomized controlled trial[pt] OR meta-analysis[pt] OR systematic review[pt] OR practice guideline[pt]). In Scopus the equivalent field-coded query was: TITLE-ABS-KEY(obesity OR overweight) AND TITLE-ABS-KEY(pharmacotherapy OR treatment OR management OR intervention OR "bariatric surgery") AND PUBYEAR > 2015 AND PUBYEAR < 2025 AND LIMIT-TO(DOCTYPE,"ar") AND LIMIT-TO(LANGUAGE,"English"). In Embase the Emtree-anchored search was: ('obesity'/exp OR 'body weight'/exp) AND ('antiobesity agent'/exp OR 'bariatric surgery'/exp OR 'diet therapy'/exp OR 'exercise'/exp) AND [2016-2024]/py AND [english]/lim AND ('randomized controlled trial'/lim OR 'meta analysis'/lim). Boolean operators were applied identically across platforms, with OR used to combine synonymous population and intervention keywords within each concept block and AND used to intersect concept blocks. The search terms were last executed on 31 December 2024.

Eligibility Criteria

Inclusion Criteria

Studies were eligible for inclusion if they met all of the following pre-specified criteria. First, study design: peer-reviewed randomised controlled trials (RCTs), systematic reviews, meta-analyses, and evidence-based clinical practice guidelines. Second, population: adults aged 18 years or older with a body mass index (BMI) of 27 kg/m² or greater; studies enrolling mixed adult-adolescent samples were eligible provided adult-stratified data were reported separately. Third, intervention: any approved or investigational pharmacological agent, structured lifestyle intervention programme, bariatric or metabolic surgical procedure, or endoscopic intervention targeting obesity. Fourth, outcomes: the primary outcome was percentage total body weight change from baseline; secondary outcomes included absolute weight loss (kg), BMI change, waist circumference reduction, HbA1c, systolic and diastolic blood pressure, major adverse cardiovascular events (MACE), and treatment-emergent adverse events. Fifth, publication characteristics:

articles published in English in journals classified as Q1 or Q2 by Scimago Journal Rank, with a CiteScore of 3.0 or higher, ensuring inclusion of only high-impact peer-reviewed work^{11,12}.

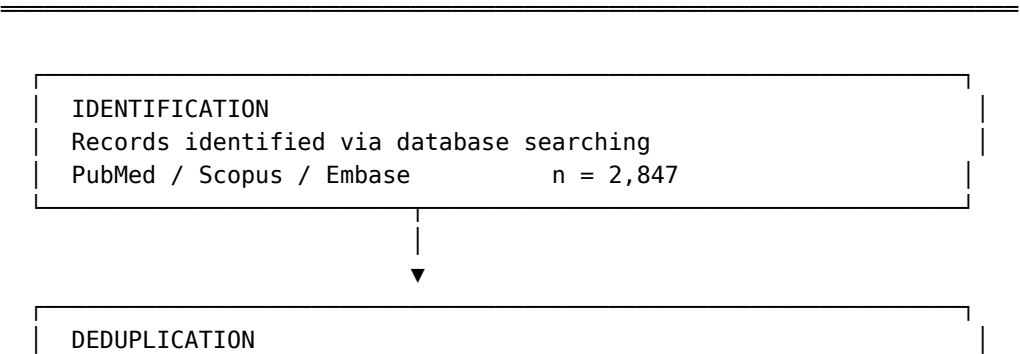
Exclusion Criteria

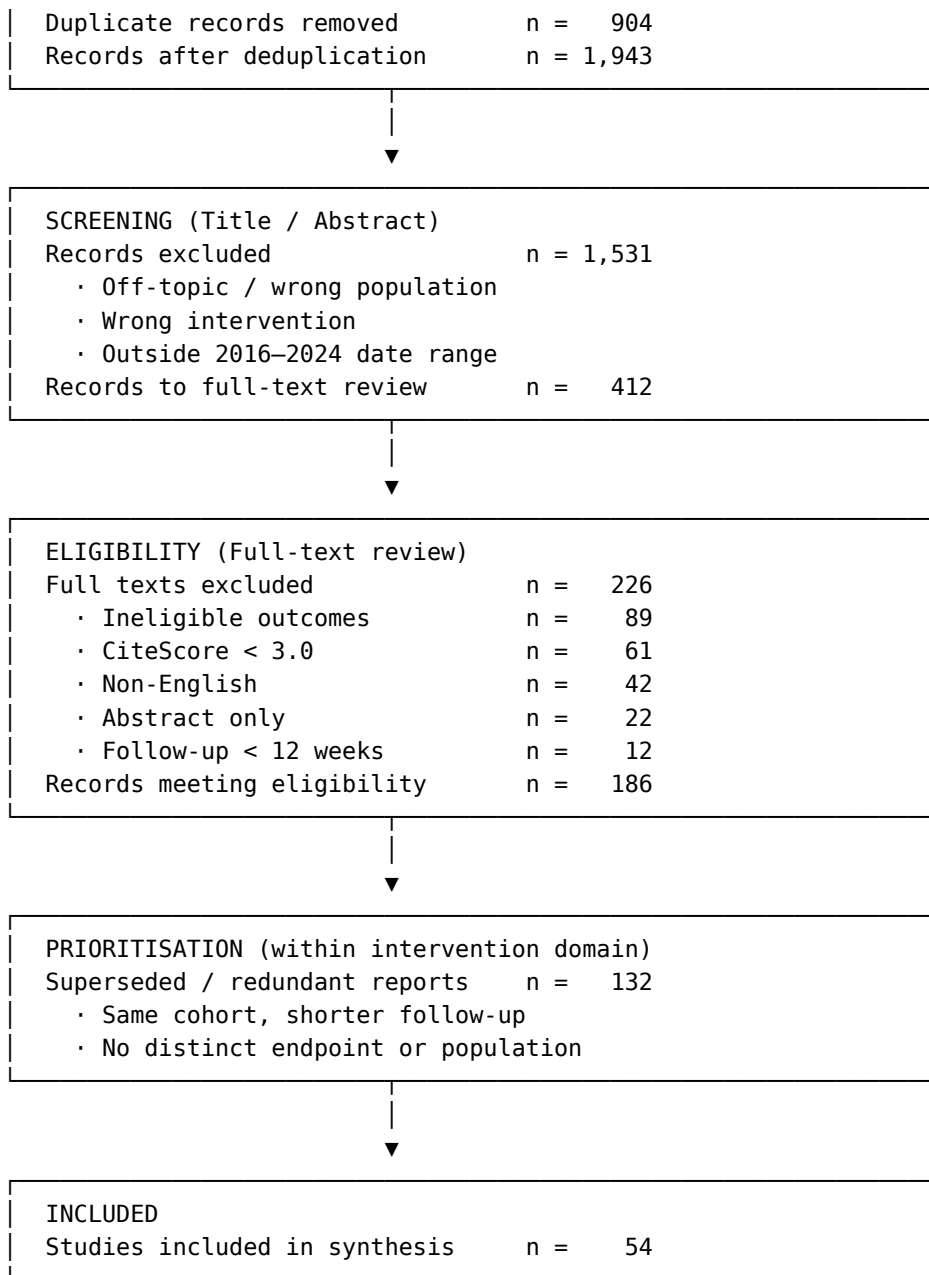
Studies were excluded if they were case reports, case series, editorials, letters, or conference abstracts without a full peer-reviewed manuscript. Research restricted to paediatric populations (age under 18 years) was excluded unless adult-disaggregated data were provided; conversely, studies specifically examining pharmacotherapy in adolescent obesity were retained as a secondary stratum^{35,36}. Articles published in languages other than English were excluded given resource constraints on translation verification. Animal studies, in vitro mechanistic studies, and purely economic modelling papers without clinical outcome data were also excluded. Studies with follow-up durations shorter than 12 weeks were considered insufficient to capture clinically meaningful weight trajectories and were excluded from the primary analysis.

PRISMA Flow

The screening and selection process followed the PRISMA 2020 four-phase flow diagram. An initial database search across PubMed, Scopus, and Embase identified 2,847 records. Following automated and manual deduplication — removing 904 records identified as duplicates across databases — 1,943 unique records proceeded to title and abstract screening. At this stage, 1,531 records were excluded because they did not meet population (BMI ≥27, adults) or intervention (pharmacological, surgical, or structured lifestyle) criteria, or were outside the 2016–2024 date range, leaving 412 records for full-text retrieval and assessment. Full-text review excluded a further 226 articles: 89 did not report eligible primary or secondary outcomes, 61 fell below the CiteScore ≥3.0 quality threshold, 42 were non-English, 22 were conference abstracts lacking full manuscripts, and 12 had follow-up durations under 12 weeks. This left 186 records meeting all eligibility criteria. Because several large trial programmes generate multiple qualifying publications reporting overlapping populations, a final prioritisation step was applied within each intervention domain: where two or more eligible reports described the same trial cohort, the publication reporting the pre-specified primary endpoint at the longest available follow-up was retained and the remainder treated as superseded. Lower-tier corroborative reports that added no distinct endpoint, population, or mechanism beyond an already-included study were likewise set aside. This removed a further 132 records and yielded a final synthesis corpus of **54 studies**.

PRISMA Flow Diagram





Quality Assessment

Risk of bias in included RCTs was assessed using the Cochrane Risk of Bias 2.0 (RoB 2.0) tool, which evaluates five domains: bias arising from the randomisation process, deviations from intended interventions, missing outcome data, measurement of the outcome, and selection of the reported result. Each domain was rated as low risk, some concerns, or high risk, with an overall judgement derived algorithmically from domain-level ratings. The SURMOUNT and STEP phase 3 programmes, which constituted the largest pharmacological RCTs in this review, were assessed as low overall risk of bias, given their double-blind design, centrally adjudicated outcomes, and

pre-registered statistical analysis plans³⁻⁵. For included systematic reviews and meta-analyses, methodological quality was appraised using the AMSTAR-2 (A Measurement Tool to Assess Systematic Reviews) instrument, which rates confidence in results across 16 domains, yielding an overall confidence rating of critically low, low, moderate, or high. Clinical practice guidelines were evaluated using the AGREE II framework^{11,12,34}.

Data Extraction

Data were extracted independently by two reviewers using a pre-piloted standardised form; discrepancies were resolved by consensus or arbitration by a third reviewer. Extracted variables included: study design, sample size, participant demographics (age, sex, baseline BMI, baseline weight), intervention details (drug name, dose, route, comparator, follow-up duration), and all eligible outcomes. Primary outcome data were extracted as mean percentage total body weight change from baseline at the primary endpoint visit, with 95% confidence intervals where available. Secondary outcomes extracted included: absolute weight loss in kilograms, BMI change (kg/m²), waist circumference change (cm), HbA1c change (% and mmol/mol), systolic blood pressure change (mmHg), incidence of MACE (defined as non-fatal myocardial infarction, non-fatal stroke, or cardiovascular death), and all-cause adverse event and serious adverse event rates. For the cardiovascular outcomes trials — LEADER³⁰, SUSTAIN-6³⁸, and SELECT⁵ — hazard ratios with 95% confidence intervals were extracted as the primary effect measure. Given the clinical and methodological heterogeneity of the included studies, data were synthesised narratively, stratified by intervention category (lifestyle, pharmacotherapy, bariatric surgery, endoscopic procedures), rather than subjected to formal meta-analytic pooling. Where multiple publications from the same trial were identified (e.g., the STEP programme for semaglutide, the SURMOUNT programme for tirzepatide), each was treated as a distinct data source reporting a pre-specified endpoint, with cross-referencing to avoid double-counting participant-level data. This data collection process — a standardized extraction form completed in duplicate, with a third-reviewer arbitration pathway — was piloted on five randomly selected studies before full deployment, and inter-reviewer agreement was monitored throughout.

Data Preparation and Synthesis

Prior to synthesis, extracted data underwent several standardized preparation steps to render findings comparable across heterogeneous reports. Weight outcomes reported only as absolute change in kilograms were converted to percentage total body weight change using the reported baseline weight, and BMI-only outcomes were retained separately rather than transformed, since back-calculation to body weight requires height data not consistently reported. HbA1c values reported in mmol/mol were converted to National Glycohemoglobin Standardization Program (NGSP) percentage units using the established linear relationship. Where trials reported both efficacy estimands (treatment-adherent) and treatment-policy estimands (intention-to-treat regardless of adherence), the treatment-policy estimand was extracted as the primary figure to preserve comparability with pragmatic effectiveness, with the efficacy estimand recorded as a secondary value. Standard errors and confidence intervals were harmonised to 95% coverage where trials reported alternative intervals.

Because the included studies differed substantially in intervention class, comparator, follow-up duration, and outcome definition, a formal meta-analysis was judged inappropriate and a structured narrative synthesis approach was adopted instead. Studies were grouped thematically by intervention domain — pathophysiological mechanism, clinical assessment, lifestyle intervention, pharmacotherapy, and surgical or procedural treatment — with each domain forming a distinct synthesis stratum. Within each stratum, findings were compared across studies by tabulating effect direction and magnitude, identifying convergent and divergent results, and explicitly noting where trial populations or endpoints precluded direct comparison. Heterogeneity was explored qualitatively through this thematic subgrouping rather than through statistical measures such as I^2 , which are not meaningful outside a pooled analysis.

Results are displayed through several complementary formats. The study selection process is presented as the PRISMA 2020 flow diagram reproduced above (Figure 1), which reports counts at each identification, screening, eligibility, and inclusion stage. Within the synthesis sections, quantitative findings are presented in the text alongside summary tables that tabulate trial name, sample size, intervention and dose, comparator, follow-up duration, and primary weight outcome, permitting visual cross-study comparison within each intervention class. Where an intervention class contained multiple dose arms — as in the SURMOUNT and STEP programmes — dose-stratified figures are reported so that dose–response relationships remain visible rather than being collapsed into a single summary estimate^{3,4}.

Reporting Bias and Certainty of Evidence

The potential for reporting bias was considered at both the study and review levels. Because formal meta-analysis was not performed, funnel plot asymmetry and Egger’s regression test could not be applied. Instead, reporting bias was assessed qualitatively by comparing published outcomes against pre-registered protocols on ClinicalTrials.gov for all included RCTs, checking for selective outcome reporting, unexplained endpoint changes, or unreported pre-specified outcomes. Publication bias remains a material concern in the obesity pharmacotherapy literature, where the large phase 3 programmes are predominantly industry-sponsored: the STEP, SURMOUNT, and SELECT trials were funded by the manufacturers of the agents under study^{3–5}. Industry sponsorship is associated with a higher likelihood of favourable published findings, and negative or null trials of investigational anti-obesity agents are less likely to reach publication. This limitation is acknowledged explicitly in the interpretation of pharmacological effect sizes and is revisited in the Discussion.

The certainty of the body of evidence supporting each major conclusion was assessed using the Grading of Recommendations, Assessment, Development and Evaluations (GRADE) framework. Evidence was rated across five domains that can lower certainty — risk of bias, inconsistency, indirectness, imprecision, and publication bias — with randomised evidence starting at high certainty and observational evidence starting at low certainty. Applying this framework, the evidence for weight reduction with GLP-1 receptor agonists and with dual GIP/GLP-1 receptor agonism was rated **high certainty**, reflecting multiple large, low-risk-of-bias, double-blind RCTs with consistent and precise effect estimates^{3,4}. The evidence for cardiovascular risk reduction with semaglutide in obesity without diabetes was likewise rated **high certainty** on the strength

of a single very large dedicated outcomes trial with adjudicated endpoints⁵. Evidence for long-term mortality benefit from bariatric surgery was rated **moderate certainty**, downgraded one level for risk of bias arising from the non-randomised, matched-cohort design of the principal supporting studies^{32,39}. Evidence for durable weight maintenance following pharmacotherapy discontinuation was rated **moderate certainty**, downgraded for imprecision given the limited number of extension studies reporting post-withdrawal trajectories^{21,40}. Evidence for emerging triple-agonist therapy was rated **low certainty**, downgraded for imprecision and indirectness given reliance on phase 2 data with short follow-up and no cardiovascular outcome data³⁷.

Pathophysiology of Obesity

Central Regulation of Energy Balance

The maintenance of body weight is governed by a tightly integrated neuroendocrine system centred on the arcuate nucleus (ARC) of the hypothalamus, which functions as the primary integrator of peripheral satiety and adiposity signals. Within the ARC, two anatomically interdigitated but functionally antagonistic neuronal populations exert opposing influences on energy homeostasis. Neuropeptide Y (NPY) and agouti-related peptide (AgRP) co-expressing neurons are potently orexigenic: when activated — as occurs during caloric deficit or low leptin states — they release NPY acting via Y1 and Y5 receptors to stimulate food intake, and AgRP, which functions as an endogenous inverse agonist at the melanocortin-4 receptor (MC4R), blocking the anorexigenic melanocortin signalling cascade. Opposing these are the pro-opiomelanocortin (POMC) and cocaine- and amphetamine-regulated transcript (CART) co-expressing neurons, which are anorexigenic: POMC undergoes post-translational cleavage by prohormone convertase 1/3 (PC1/3) to yield α -melanocyte-stimulating hormone (α -MSH), which binds MC4R in the paraventricular nucleus (PVN) to reduce food intake and increase energy expenditure^{14,18}.

MC4R occupies a position of singular importance in human obesity genetics. Loss-of-function variants in the *MC4R* gene are identified in approximately 2–6% of individuals with severe obesity, constituting the most common monogenic cause of common obesity¹⁸. The receptor signals primarily via G α s-coupled adenylyl cyclase activation, but also recruits β -arrestin pathways that modulate receptor internalisation and downstream MAPK signalling. The upstream regulator of this axis is leptin, a 16-kDa adipokine encoded by the *LEP* gene and secreted proportionally to adipose tissue mass. Leptin traverses the blood–brain barrier via a saturable transport mechanism and binds the long-form leptin receptor (LEPRb) on ARC POMC neurons, activating JAK2–STAT3 phosphorylation and driving *Pomc* gene transcription²⁰. In the obese state, despite dramatically elevated circulating leptin concentrations — commonly exceeding 30–40 ng/mL compared with 5–10 ng/mL in lean individuals — central leptin resistance emerges through multiple mechanisms including downregulation of LEPRb expression, impaired JAK2 signalling secondary to suppressor of cytokine signalling-3 (SOCS3) induction, and reduced leptin transport across the blood–brain barrier. This functional leptin resistance effectively uncouples adiposity from the homeostatic braking mechanisms that should limit further weight gain, creating a permissive neurobiological environment for progressive obesity.

The Gut–Brain Axis: Incretin and Satiety Hormones

The gastrointestinal tract functions as the body's largest endocrine organ and is an essential afferent limb of the gut–brain energy-sensing axis. Glucagon-like peptide-1 (GLP-1) is derived from the *GCG* gene product proglucagon through differential post-translational processing in intestinal L-cells concentrated in the distal ileum and colon; in contrast, pancreatic α -cells process the same proglucagon precursor to yield glucagon^{16,41}. Following nutrient ingestion, L-cells secrete GLP-1 into the portal circulation within minutes; GLP-1 then acts through multiple mechanisms to reduce energy intake: it slows gastric emptying, augments glucose-stimulated insulin secretion from pancreatic β -cells, suppresses glucagon secretion from α -cells, and — crucially for obesity pharmacotherapy — activates GLP-1 receptors (GLP-1R) on vagal afferent neurons innervating the hepatoportal region and nodose ganglion, transmitting satiety signals to the nucleus tractus solitarius (NTS) in the brainstem, which in turn relays these signals to hypothalamic centres¹⁷. GLP-1R are also expressed directly on ARC neurons, providing a direct central pathway for GLP-1-mediated appetite suppression.

Glucose-dependent insulintropic polypeptide (GIP), secreted from K-cells of the proximal small intestine in response to fat and carbohydrate ingestion, was originally characterised for its insulintropic properties but has more recently been recognised as an important regulator of adipose tissue metabolism; GIP receptor (GIPR) signalling in adipocytes promotes lipid storage under fed conditions and may modulate energy expenditure, and in the central nervous system GIPR is expressed in the hypothalamus and ventral tegmental area, suggesting roles in reward-related eating behaviour⁴². The synergistic combination of GLP-1R and GIPR agonism, as achieved by tirzepatide, produces greater weight loss than selective GLP-1R agonism alone — a finding with profound mechanistic implications. Peptide YY (PYY), co-secreted with GLP-1 from L-cells, acts on Y2 receptors in the ARC to inhibit NPY/AgRP neurons and thereby suppress appetite. Cholecystokinin (CCK), released from I-cells of the duodenum and jejunum in response to dietary fat and protein, activates CCK-A receptors on vagal afferents and mediates short-term meal termination; unlike leptin and insulin, CCK resistance does not appear to develop in obesity, making its pathway a continued target of pharmacological interest¹⁸.

Adipose Tissue as an Endocrine Organ

White adipose tissue (WAT) is no longer regarded as a passive energy reservoir but as a metabolically active endocrine organ that secretes over 600 bioactive proteins collectively termed adipokines. The contrast between visceral adipose tissue (VAT), concentrated in the omental and mesenteric depots, and subcutaneous adipose tissue (SAT) is of central clinical significance: VAT is characterised by greater lipolytic activity, higher macrophage infiltration, and proximity to the portal circulation, resulting in delivery of free fatty acids and pro-inflammatory cytokines directly to the liver, whereas SAT acts as a relatively protective buffer depot with lower inflammatory tone^{14,20}. Leptin, the archetypal adipokine, is produced in direct proportion to adipocyte lipid content and fat mass, yet fails functionally in the obese hypothalamus as described above. Adiponectin, encoded by *ADIPOQ*, exhibits the opposite pattern: its circulating levels fall progressively with increasing adiposity and insulin resistance, a paradox partly explained by the fact that adiponectin has insulin-sensitising and anti-inflammatory properties — signalling

through AdipoR1 and AdipoR2 receptors to activate AMPK and PPAR- α pathways — that are lost when adiponectin secretion is impaired. Resistin, interleukin-6 (IL-6), and tumour necrosis factor- α (TNF α) are pro-inflammatory adipokines that increase with visceral adiposity; TNF α impairs insulin signalling through serine phosphorylation of insulin receptor substrate-1 (IRS-1), contributing to peripheral insulin resistance and a chronic low-grade inflammatory state that further drives adipose tissue dysfunction and cardiometabolic risk²⁰.

The Gut Microbiome and Metabolic Crosstalk

A landmark contribution to understanding obesity pathophysiology came from¹⁹, who demonstrated in a germ-free mouse model that colonisation with caecal microbiota harvested from obese donors produced a significantly greater increase in total body fat than colonisation with microbiota from lean donors — without any difference in caloric intake — establishing a causal role for the gut microbiome in energy harvest and fat deposition. Metagenomic studies in human cohorts have consistently reported an elevated Firmicutes-to-Bacteroidetes ratio in individuals with obesity, although the mechanistic interpretation remains nuanced, as functional gene capacity for polysaccharide fermentation appears more important than phylum-level composition. The microbiome influences host energy metabolism through several interconnected mechanisms: production of short-chain fatty acids (SCFAs) — principally acetate, propionate, and butyrate — from dietary fibre fermentation, which signal through G-protein-coupled receptors GPR41 (FFAR3) and GPR43 (FFAR2) on colonocytes, enteroendocrine L-cells, and adipocytes, stimulating PYY and GLP-1 secretion; bile acid biotransformation, whereby primary bile acids undergo 7 α -dehydroxylation by Clostridiales to yield secondary bile acids that activate Takeda G-protein receptor 5 (TGR5) on L-cells, further stimulating GLP-1 release; and regulation of intestinal permeability and endotoxaemia through the toll-like receptor 4 (TLR4)-lipopolysaccharide axis, which contributes to metabolic inflammation¹⁹.

Genetic Architecture and Developmental Programming

The heritability of BMI is estimated at 40–70% from twin studies, reflecting both polygenic and rare monogenic contributions. The fat mass and obesity-associated gene (*FTO*) represents the most replicated common variant locus: the rs9939609 risk allele, present in approximately 16% of individuals of European ancestry, is associated with an average 1.5 kg increase in body weight and a 0.4 kg/m² increment in BMI per allele, acting partly through a regulatory role on *IRX3* and *IRX5* expression in adipocyte precursors that shifts the balance from thermogenic beige adipocyte differentiation toward lipid-storing white adipocyte differentiation^{14,18}. Beyond *FTO*, genome-wide association studies have identified over 900 loci contributing to BMI variance. At the monogenic extreme, complete leptin deficiency arising from homozygous *LEP* mutations produces severe early-onset obesity that is fully reversible with recombinant leptin replacement — a proof-of-concept demonstration of the leptin–melanocortin axis’s centrality. Leptin receptor (*LEPR*) mutations produce a phenotypically similar syndrome with additional hyperpituitarism. Epigenetic mechanisms, including DNA methylation changes at developmental programming windows — particularly gestational overnutrition and excess gestational weight gain — imprint adipogenic gene expression patterns that persist through life, linking maternal metabolic health

to offspring obesity risk via mechanisms operating through the hypothalamic–pituitary– adrenal (HPA) axis and altered glucocorticoid receptor methylation.

Obesogenic Environmental Drivers

The biological predisposition to obesity is amplified by pervasive environmental exposures that systematically disrupt homeostatic energy regulation. Ultra-processed foods, classified under the NOVA system as industrially formulated products containing ingredients and additives not used in domestic cooking, are independently associated with excess energy intake beyond that explained by their nutrient profiles, driven partly by engineered hyperpalatability operating through mesolimbic dopamine circuits and partly by rapid gastric transit that attenuates post-meal GLP-1 and PYY secretion⁴³. Sleep curtailment to below seven hours per night reliably suppresses leptin and elevates acyl-ghrelin – the orexigenic stomach-derived peptide – creating a hormonal milieu that increases appetite by an estimated 500 kcal/day. Chronic psychosocial stress activates the HPA axis, elevating cortisol and promoting preferential VAT deposition through glucocorticoid receptor-mediated lipid uptake in visceral adipocytes. The built environment, characterised by food deserts, limited active transport infrastructure, and advertising saturation of energy-dense products, structures behavioural choices in ways that are effectively invisible to individual volition. Finally, endocrine-disrupting chemicals including bisphenol A (BPA) and phthalates – found in food packaging and consumer products – interfere with adipogenesis and thyroid hormone signalling, contributing to the so-called “chemical obesogens” hypothesis¹⁴.

Pharmacotherapeutic Exploitation of Pathophysiological Pathways

The foregoing mechanistic landscape directly defines the target space for anti-obesity pharmacotherapy. GLP-1R agonists such as semaglutide exploit the gut–brain satiety axis by providing sustained, pharmacological-level GLP-1R stimulation that exceeds the amplitude and duration achievable by endogenous post-prandial secretion, thereby engaging both vagal afferent and direct hypothalamic POMC-neuron pathways to reduce appetite and food reward^{3,16}. Tirzepatide adds GIPR co-agonism to this framework; the synergistic weight loss produced – up to 22.5% of body weight at the 15 mg dose in SURMOUNT-1 – likely reflects additive or supra-additive engagement of complementary hypothalamic and adipocyte-level circuits⁴. Naltrexone–bupropion targets the melanocortin pathway directly: bupropion activates POMC neurons and increases α -MSH release, while naltrexone blocks β -endorphin-mediated autoinhibitory feedback on POMC neurons, resulting in sustained melanocortin tone²⁷. Orlistat operates peripherally, inhibiting intestinal lipase to reduce dietary fat absorption by approximately 30%, increasing the luminal fat load available to stimulate L-cell GLP-1 secretion as a secondary benefit. Bariatric surgery exploits multiple overlapping pathways simultaneously – caloric restriction, altered bile acid–TGR5 signalling, microbiome remodelling, and dramatic post-prandial GLP-1 surges – that together account for weight loss outcomes substantially exceeding those attributable to caloric restriction alone⁴⁴. The convergence of mechanistic understanding thus frames pharmacotherapy not as a blunt metabolic suppressor but as a precision modulator of the specific neuroendocrine and peripheral circuits whose dysregulation drives the pathophysiology of obesity.

Assessment and Classification of Obesity

The accurate assessment of obesity is a prerequisite for appropriate clinical decision-making, yet the tools available to clinicians vary substantially in their complexity, accessibility, and validity. Body mass index (BMI), calculated as weight in kilograms divided by the square of height in metres, remains the most widely used metric for population-level and clinical classification of adiposity. The World Health Organization categorises BMI into underweight (below 18.5 kg/m²), normal weight (18.5–24.9 kg/m²), overweight (25.0–29.9 kg/m²), and obesity (30.0 kg/m² and above), with obesity further subdivided into class I (30.0–34.9 kg/m²), class II (35.0–39.9 kg/m²), and class III, or severe obesity (40.0 kg/m² and above). These thresholds underpin treatment eligibility criteria across pharmacological, endoscopic, and surgical domains and are enshrined in clinical practice guidelines internationally¹². Recognising that Asian populations accrue cardiometabolic risk at lower thresholds of adiposity, the WHO Expert Consultation proposed revised Asia-Pacific cut-points: overweight at a BMI of 23.0 kg/m² or above and obesity at 27.5 kg/m² or above. These ethnicity-specific thresholds are now integrated into guidelines across East and South Asia and are increasingly advocated in multiethnic Western clinical settings to avoid systematic underdiagnosis in these populations⁴³.

Despite its administrative convenience, BMI is an imprecise surrogate for adiposity that carries several well-characterised limitations. Because BMI is derived from total body mass relative to height, it cannot distinguish fat mass from lean mass; muscular individuals and those with sarcopenia may therefore be misclassified. Equally important, BMI provides no information about the anatomical distribution of adipose tissue, a distinction of considerable physiological consequence. Two individuals with identical BMI values may differ dramatically in cardiometabolic risk depending on the relative proportions of visceral and subcutaneous fat. The mortality data are also more nuanced than the WHO classification suggests: the landmark meta-analysis by¹³, pooling data from 97 prospective studies and more than 2.88 million individuals, found that grade 1 obesity (BMI 30.0–34.9 kg/m²) was not associated with significantly increased all-cause mortality compared with normal weight, whereas grades II and III combined were associated with a hazard ratio of 1.29 (95% CI 1.18–1.41). This finding, while contested methodologically, underscores that BMI alone is insufficient to stratify individual risk¹³.

The limitations of BMI have stimulated interest in measures that better capture the distribution and health consequences of adiposity. Waist circumference is the most clinically practical of these measures. The National Cholesterol Education Program Adult Treatment Panel III (NCEP-ATP III) defines abdominal obesity as waist circumference exceeding 102 cm in men and 88 cm in women, thresholds associated with substantially elevated risk of metabolic syndrome and cardiovascular disease. As with BMI, ethnicity-specific thresholds are recognised: the International Diabetes Federation uses 90 cm in Asian men and 80 cm in Asian women. The waist-to-height ratio — ideally maintained below 0.5 across adulthood — combines information about central fat distribution with an implicit adjustment for stature and has demonstrated predictive validity for cardiovascular and metabolic outcomes across diverse populations⁴⁵. Where greater precision is required, dual-energy X-ray absorptiometry (DXA) serves as the clinical gold standard for body composition assessment, enabling direct quantification of fat mass, lean mass, and visceral adipose

tissue, and is particularly valuable in the assessment of sarcopenic obesity and in the evaluation of treatment response in clinical trials.

The Edmonton Obesity Staging System (EOSS), proposed by¹⁵, represents a substantive advance beyond purely anthropometric classification by incorporating the functional and clinical consequences of obesity. The EOSS assigns patients to one of five stages based on the presence and severity of metabolic, physical, and mental health complications. Stage 0 denotes no apparent risk factors, functional limitations, or psychological symptoms despite excess adiposity; stage 1 indicates subclinical risk factors such as borderline hypertension or fasting glucose impairment; stage 2 encompasses established comorbidities including type 2 diabetes, hypertension, obstructive sleep apnoea, or moderate functional limitations; stage 3 characterises severe organ damage, significant disability, or major psychiatric illness; and stage 4 reflects end-stage organ failure or extreme functional incapacity. The clinical utility of this staging scheme is that it shifts the treatment imperative away from BMI alone and toward the burden of obesity-related illness. Importantly, epidemiological analyses have demonstrated that EOSS predicts all-cause mortality independently of and more accurately than BMI, challenging the notion that weight reduction should be pursued uniformly across all individuals with elevated BMI^{12,15}.

Building on similar principles, the Adiposity-Based Chronic Disease (ABCD) framework was proposed as a diagnostic and staging framework that emphasises complications as the primary driver of clinical action. Within the ABCD construct, abnormal adiposity constitutes the primary diagnosis, while the nature and severity of metabolic (such as dysglycaemia, dyslipidaemia, and hypertension), biomechanical (such as osteoarthritis and sleep-disordered breathing), and psychological (such as depression and binge eating disorder) complications determine staging and guide treatment intensity. The explicit complications-driven orientation of the ABCD framework represents a movement away from weight-centric medicine toward a model in which cardiometabolic and functional health outcomes, rather than weight per se, anchor the therapeutic rationale¹².

A comprehensive metabolic risk assessment is therefore integral to the obesity evaluation. Metabolic syndrome is diagnosed according to either the IDF or NCEP-ATP III criteria when three or more of the following are present: elevated waist circumference (using ethnicity-appropriate thresholds), fasting plasma glucose at or above 5.6 mmol/L (100 mg/dL) or established type 2 diabetes, serum triglycerides at or above 1.7 mmol/L (150 mg/dL), HDL-cholesterol below 1.0 mmol/L (40 mg/dL) in men or below 1.3 mmol/L (50 mg/dL) in women, and blood pressure at or above 130/85 mmHg or pharmacological treatment for hypertension. Beyond these components, the clinical assessment of obesity should incorporate HbA1c and fasting glucose for glycaemic stratification, a full fasting lipid profile, and a cardiovascular risk calculator such as the Pooled Cohort Equations or SCORE2, which integrate multiple risk factors into absolute 10-year cardiovascular event probabilities¹¹. Given the high prevalence of metabolic-associated steatotic liver disease in this population, hepatic evaluation using non-invasive indices such as the Fibrosis-4 (FIB-4) score and liver stiffness measurement by elastography is increasingly recommended as part of the initial workup.

Assessment must also account for clinically important variations in the obesity phenotype that standard metrics fail to capture. Sarcopenic obesity — defined by the coexistence of excess adi-

posity and reduced muscle mass or strength — is increasingly prevalent in older adults and confers risks not adequately reflected in BMI-based classifications; it is best identified through DXA-based body composition measurement combined with grip strength or functional performance testing. Conversely, so-called normal-weight metabolic obesity describes individuals whose BMI falls within the normal range yet who harbour excess visceral adiposity and a markedly adverse cardiometabolic risk profile, further illustrating the limitations of weight-based assessment. The obesity paradox — observed associations between overweight or grade 1 obesity and improved survival in certain chronic diseases including heart failure and chronic kidney disease — complicates treatment decisions in these populations and may reflect, at least in part, methodological confounding by reverse causation and BMI's insensitivity to body composition. Finally, weight assessment in pregnancy requires specific consideration: gestational weight gain recommendations from the Institute of Medicine are stratified by pre-pregnancy BMI and reflect the dual imperative of supporting fetal development while minimising the risk of gestational diabetes, preeclampsia, and excessive postpartum weight retention. Taken together, these considerations affirm that a clinically meaningful assessment of obesity must integrate anthropometric measurements, body composition data, the full cardiometabolic risk profile, and the broader functional and psychological context of the individual patient.

Lifestyle Interventions

Lifestyle modification — encompassing dietary change, increased physical activity, and behavioural therapy — constitutes the cornerstone of obesity management and a necessary adjunct to all pharmacological and surgical treatments. The evidence base for these interventions spans decades and encompasses landmark randomised trials, large pragmatic studies, and meta-analyses that collectively delineate both the substantial short-term efficacy and the well-recognised long-term challenges of lifestyle approaches to weight reduction.

The most fundamental dietary principle in weight management is the creation of a sustained energy deficit. A deficit of 500–1,000 kcal per day relative to estimated total energy expenditure is projected to yield a weight loss of approximately 0.5–1.0 kg per week, a target widely endorsed by clinical practice guidelines⁴⁶. Crucially, the macronutrient composition of the energy-restricted diet — whether low fat, low carbohydrate, high protein, or Mediterranean in pattern — appears to exert less influence on total weight loss than the degree to which the individual adheres to the prescribed regimen, a conclusion supported by head-to-head comparative trials and systematic reviews. For patients with class II or III obesity in whom more rapid weight reduction is clinically indicated, very-low-calorie diets (VLCDs), providing fewer than 800 kcal per day, typically through meal-replacement formulas, can achieve losses of 1.5–2.5 kg per week. VLCDs require close medical supervision and often employ a protein-sparing modified fast design to minimise lean mass catabolism, but are associated with improved adherence to subsequent weight maintenance programmes when combined with structured behavioural support²⁸. Specific dietary patterns carry additional evidence-based benefits beyond weight reduction. The PREDIMED trial demonstrated that a Mediterranean dietary pattern, characterised by abundant olive oil, nuts, legumes, fish, and modest wine consumption, significantly reduced the composite incidence of major cardiovascular events among high-risk adults compared with a low-fat control diet, inde-

pendent of modest differences in achieved weight. Low-carbohydrate and ketogenic diets, which restrict carbohydrate intake to below approximately 50 g per day, consistently produce superior weight loss at 6 months relative to isocaloric low-fat diets, with more favourable reductions in triglycerides and increases in HDL-cholesterol; however, these differences largely attenuate by 12 months as dietary adherence converges, suggesting that macronutrient manipulation provides no durable metabolic advantage beyond what is attributable to the energy deficit itself⁴⁶. Intermittent fasting strategies – including time-restricted eating (typically an 8-hour eating window), the 5:2 approach (severe restriction on two non-consecutive days per week), and alternate-day fasting – have accumulated a meaningful evidence base demonstrating weight loss comparable to continuous caloric restriction over equivalent follow-up periods. Emerging data suggest that some metabolic benefits of time-restricted eating, including improvements in insulin sensitivity and circadian alignment of nutrient timing, may operate through pathways partially independent of total caloric intake, though the clinical significance of these effects remains under investigation.

Physical activity is a second essential pillar of lifestyle management, with distinct roles in weight loss, cardiometabolic risk reduction, and the prevention of weight regain. The World Health Organization recommends that adults accumulate at least 150–300 minutes of moderate-intensity aerobic activity – or at least 75–150 minutes of vigorous-intensity activity, or an equivalent combination – per week, alongside muscle-strengthening activities on two or more days⁴⁵. Aerobic exercise alone, without dietary restriction, typically produces modest weight loss of approximately 2–3 kg over 6 months in clinical trials, a figure substantially less than what is achieved through dietary change alone, likely because voluntary food intake tends to partially compensate for exercise-induced energy expenditure. However, this understates the importance of physical activity, both because of its broad cardiometabolic benefits independent of weight change – including reductions in hepatic fat, improvement in insulin sensitivity, and favourable effects on blood pressure and lipid profiles – and because of its critical role in long-term weight maintenance. Resistance training, while producing minimal weight loss in isolation, preserves lean mass during phases of dietary energy restriction, thereby protecting against the reduction in resting metabolic rate that accompanies loss of skeletal muscle; for this reason, a combined aerobic and resistance training regimen is superior to either modality alone in the context of obesity treatment. The Diabetes Prevention Program demonstrated that participants randomised to intensive lifestyle intervention who achieved the physical activity goal of at least 150 minutes of moderate-intensity walking per week maintained a mean weight loss of 4.9 kg at three years, illustrating the particular salience of physical activity for weight maintenance even when initial weight loss is achieved primarily through dietary means²².

Behavioural interventions provide the cognitive and motivational scaffolding through which dietary and physical activity goals are achieved and sustained. Motivational interviewing, cognitive-behavioural therapy (CBT), and acceptance and commitment therapy (ACT) address, respectively, intrinsic motivation, maladaptive eating cognitions, and psychological flexibility. Among specific behavioural techniques, self-monitoring of dietary intake and physical activity consistently emerges as the strongest independent predictor of weight maintenance; digital tools including smartphone applications and wearable trackers amplify this effect when integrated into structured programmes⁴⁶. Structured commercial programmes such as Weight Watchers and

Jenny Craig have produced clinically meaningful results in controlled trials. The landmark Look AHEAD trial provides the most comprehensive evidence base for intensive lifestyle intervention in obesity with comorbid diabetes: among 5,145 adults randomised to intensive intervention versus standard diabetes education, the intervention group achieved a mean weight loss of 8.6% of initial body weight at 1 year, sustained at a mean of 4.7% at 4 years, with concurrent improvements in HbA1c, blood pressure, and lipids²⁴. Critically, however, the trial was stopped early because it failed to demonstrate a reduction in its primary composite cardiovascular endpoint — nonfatal myocardial infarction, nonfatal stroke, hospitalisation for angina, or cardiovascular death — over a median follow-up of 9.6 years, distinguishing cardiometabolic risk factor improvement from hard event prevention.

The two foundational randomised trials of lifestyle-based diabetes prevention remain the most cited evidence for intensive lifestyle intervention in high-risk populations. The Finnish DPS, reported by²³, randomised 522 overweight adults with impaired glucose tolerance to a five-component lifestyle programme targeting more than 5% weight loss, reduced saturated fat, increased fibre, and at least 30 minutes of daily moderate activity; after a mean follow-up of 3.2 years, type 2 diabetes incidence fell by 58%. The DPP, reported by²², randomised 3,234 high-risk adults to intensive lifestyle intervention, metformin 850 mg twice daily, or placebo; at 2.8 years the lifestyle arm achieved a 58% reduction in diabetes incidence versus placebo, compared with 31% for metformin, with the lifestyle benefit driven by a mean weight loss of 5.6 kg and consistent attainment of the 150-minute weekly walking goal.

Despite these encouraging findings, realistic appraisal of lifestyle interventions requires acknowledgement of their biological limitations. The weight loss plateau, almost universally observed between 6 and 12 months, reflects not only faltering adherence but powerful counter-regulatory adaptations: adaptive thermogenesis reduces resting metabolic rate by approximately 12–15% below the level predicted for the new body weight after losses of only 6–8% of initial mass, while suppression of leptin and elevation of ghrelin sustain the homeostatic drive toward weight regain for years¹⁴. Without continued intervention, most individuals regain at least one-third of lost weight within one year, and the majority return to baseline within five years. These findings argue for framing lifestyle modification not as a time-limited course but as a sustained component of a longitudinal management strategy, integrated with pharmacotherapy or surgery where lifestyle alone proves insufficient^{45,46}.

Pharmacotherapy for Obesity

Historical Context and FDA-Approved Agents

The pharmacological treatment of obesity has been punctuated by the withdrawal of agents whose risks outweighed their benefits. Amphetamine derivatives dominated mid-twentieth-century practice, followed by the fenfluramine–phentermine combination withdrawn in 1997 after reports of valvular heart disease. The endocannabinoid receptor antagonist rimonabant was withdrawn globally by 2008 following evidence of severe neuropsychiatric adverse events, including suicidality. The norepinephrine–serotonin reuptake inhibitor sibutramine was removed from markets in 2010 after the Sibutramine Cardiovascular OUTcomes (SCOUT) trial, enrolling 10,742 high-

risk patients, demonstrated a significant increase in nonfatal myocardial infarction and nonfatal stroke compared with placebo²⁸. These withdrawals established a stringent safety framework and reinforced the primacy of cardiovascular outcomes data for new agents, prompting the FDA to require dedicated cardiovascular outcomes trials for all new anti-obesity medications, a policy that has since generated pivotal data redefining the risk–benefit calculus of pharmacological weight management. As of 2024, six agents or combinations hold FDA approval for chronic weight management: orlistat, phentermine/topiramate extended-release, naltrexone/bupropion extended-release, liraglutide 3.0 mg daily, semaglutide 2.4 mg weekly, and tirzepatide 5–15 mg weekly (approved November 2023). The magnitude of pharmacologically achievable weight loss has expanded from approximately 3–4 kg with orlistat to more than 20% of body weight with tirzepatide, a trajectory reflecting deepening mechanistic insight into hypothalamic and peripheral energy regulation. Clinical practice guidelines from the Endocrine Society and the American Association of Clinical Endocrinologists provide structured frameworks for agent selection based on comorbidities, contraindications, and patient preference^{11,12}.

Older Anti-Obesity Medications

Orlistat (Xenical) 120 mg three times daily acts as a reversible inhibitor of pancreatic lipase, reducing dietary fat absorption by approximately 30%. The landmark XENical in the Prevention of Diabetes in Obese Subjects (XENDOS) trial, reported by²⁵ in 3,305 obese Swedish adults over four years, demonstrated a 37% relative risk reduction in type 2 diabetes incidence compared with placebo (6.2% versus 9.0%, $p = 0.0032$), alongside a 2.8 kg incremental weight loss advantage. Gastrointestinal adverse effects—oily spotting, fecal urgency, and fecal incontinence in up to 27% of users—substantially limit long-term adherence, though the absence of systemic absorption and cardiovascular risk makes orlistat suitable when other agents are contraindicated.

The combination of phentermine and topiramate extended-release (Qsymia) exploits complementary mechanisms: catecholaminergic appetite suppression from phentermine and GABAergic satiety enhancement from topiramate. In the pivotal CONQUER trial,²⁶ enrolled 2,487 overweight or obese adults with two or more comorbidities and demonstrated weight losses of 7.8% (7.5 mg/46 mg dose) and 9.8% (15 mg/92 mg dose) of baseline body weight versus 1.2% with placebo at 56 weeks ($p < 0.001$ for both), accompanied by reductions in blood pressure and fasting glucose. Topiramate's teratogenicity—conferring an approximately five-fold elevated risk of oral cleft with first-trimester exposure—necessitates a mandatory REMS program with monthly pregnancy testing in women of childbearing potential.

The fixed-dose combination of naltrexone 32 mg and bupropion 360 mg extended-release (Contrave) engages the hypothalamic melanocortin system: bupropion activates arcuate nucleus pro-opiomelanocortin (POMC) neurons, increasing alpha-melanocyte-stimulating hormone, while naltrexone blocks mu-opioid autoinhibitory feedback on POMC neurons to sustain anorectic signaling. The COR-I trial, conducted by²⁷ in 1,453 overweight and obese adults, demonstrated 6.1% weight loss versus 1.3% with placebo at 56 weeks ($p < 0.001$), with 42% versus 17% achieving at least 5% weight loss. The combination carries boxed warnings for neuropsychiatric adverse events and is contraindicated in patients receiving opioid analgesics.

GLP-1 Receptor Agonists

GLP-1 receptor agonists act on GLP-1 receptors distributed throughout the hypothalamic arcuate nucleus, the nucleus tractus solitarius, and vagal afferent neurons to suppress appetite, promote satiety, and delay gastric emptying, while peripheral receptor activation in the pancreas augments glucose-dependent insulin secretion^{16,41}. Liraglutide 3.0 mg, administered as a daily subcutaneous injection, was evaluated in the SCALE Obesity and Prediabetes trial by²⁹, which randomized 3,731 non-diabetic adults to liraglutide or placebo with lifestyle counseling for 56 weeks. The liraglutide group achieved a mean weight loss of 8.0% versus 2.6% with placebo ($p < 0.001$), with 63.2% versus 27.1% achieving at least 5% weight loss. Nausea occurred in approximately 40% and vomiting in approximately 15% of participants, predominantly during the first eight weeks of dose escalation.

Semaglutide 2.4 mg weekly, engineered with an extended half-life enabling greater receptor occupancy than liraglutide, produced transformative results in the STEP program. The STEP 1 trial, reported by³ in 1,961 non-diabetic adults (mean body weight 105.4 kg), achieved a mean weight loss of 14.9% versus 2.4% with placebo at 68 weeks ($p < 0.001$), corresponding to an absolute loss of 15.3 kg; 86.4% versus 31.5% achieved at least 5% weight loss, and 50.5% versus 4.9% achieved at least 15% weight loss. In the STEP 2 trial among 1,210 adults with type 2 diabetes,⁴⁷ demonstrated 9.6% versus 3.4% weight loss ($p < 0.001$), with attenuation attributable to impaired incretin responsiveness in established insulin resistance. The STEP 3 trial, reported by⁴⁸ in 611 participants receiving semaglutide as an adjunct to intensive behavioral therapy with a structured low-calorie phase, produced 16.0% versus 5.7% weight loss, establishing the additive value of combined pharmacological and behavioral intervention. A direct head-to-head comparison by⁴⁹ confirmed superiority of semaglutide over liraglutide for body weight reduction. Critically, weight regain after cessation is substantial: the STEP 1 extension reported by²¹ demonstrated recovery of approximately two-thirds of prior weight loss within one year of drug withdrawal, establishing the requirement for indefinite treatment continuation.

The cardiovascular significance of semaglutide 2.4 mg was definitively established by the SELECT trial, reported by⁵, which enrolled 17,604 adults with established cardiovascular disease and obesity but without diabetes, following them for a median of 3.3 years. Semaglutide produced a 20% relative risk reduction in the primary composite endpoint of cardiovascular death, nonfatal myocardial infarction, or nonfatal stroke (HR 0.80, 95% CI 0.72–0.90, $p < 0.001$), and this benefit was independent of the degree of weight loss achieved, suggesting direct cardioprotective effects of GLP-1R agonism beyond weight and glucose lowering.

Dual GIP/GLP-1 Receptor Agonist: Tirzepatide

Tirzepatide is a synthetic dual agonist at both the GIP receptor and GLP-1R. GIP receptor activation in adipose tissue amplifies lipolysis and reduces lipotoxicity, while synergistic hypothalamic signaling through both receptors confers greater anorectic potency than pure GLP-1R agonism^{18,42}. The SURMOUNT-1 trial, reported by⁴ in 2,539 non-diabetic adults, demonstrated dose-dependent weight losses of 15.0% (5 mg), 19.5% (10 mg), and 20.9% (15 mg) versus 3.1% with placebo at 72 weeks ($p < 0.001$ for all); 57.9% of the 15 mg group achieved at least 20% weight loss. Nausea was reported by 39.4% of 15 mg participants, predominantly mild to moderate and transient during dose escalation. Among 938 adults with type 2 diabetes in SURMOUNT-2,⁵⁰ reported weight losses

of 12.8%, 14.7%, and 15.7% with tirzepatide 5 mg, 10 mg, and 15 mg versus 3.3% with placebo ($p < 0.001$). In SURMOUNT-3,⁵¹ demonstrated that tirzepatide following an intensive lifestyle lead-in yielded 18.4% weight loss from randomization versus 2.5% with placebo in 806 participants. Maintenance was examined by⁴⁰ in SURMOUNT-4, where 670 participants completing 36 weeks of open-label tirzepatide (mean -20.9% body weight) were randomized to continue or switch to placebo; continuing tirzepatide produced a further -3.4 kg versus $+11.2$ kg with placebo switch ($p < 0.001$), again confirming the need for ongoing treatment. Tirzepatide also demonstrated potent efficacy for metabolic dysfunction-associated steatohepatitis (MASH) in the SURMOUNT-NASH trial, reported by⁶ in 190 adults with biopsy-confirmed MASH and F2–F3 fibrosis; MASH resolution without fibrosis worsening occurred in 62.4% (10 mg) and 79.1% (15 mg) versus 9.7% with placebo, and fibrosis improvement by at least one stage in 35.4% and 44.9% versus 14.9% ($p < 0.001$ for both).

Emerging Pharmacotherapy

The investigational pipeline has produced efficacy signals that exceed those of any approved agent. Retatrutide, a triple agonist at GLP-1R, GIPR, and the glucagon receptor, exploits glucagon receptor stimulation to enhance hepatic fatty acid oxidation and thermogenesis in addition to incretin-mediated appetite suppression. In a phase 2 trial reported by³⁷ enrolling 338 adults over 48 weeks, the 12 mg dose produced a mean body weight reduction of 24.2%—the largest pharmacologically achieved weight loss in a randomized trial as of 2024. Amylin analogue combinations (cagrilintide plus semaglutide, or CagriSema) and anti-ActRII monoclonal antibodies such as bimagrumab, which preferentially reduces fat while preserving lean mass, represent further mechanistic diversification. The broader pipeline reviewed by¹⁸ encompasses FGF21 analogues, mitochondrial uncouplers, and PCSK9-targeted approaches, collectively presaging an era of combination pharmacotherapy targeting multiple energy-regulatory axes simultaneously.

Practical Considerations

Shared decision-making should integrate patient preferences, cost, insurance coverage, contraindications, and comorbidity profile. Adverse events common to GLP-1R and GIP/GLP-1R agonists—nausea, vomiting, and diarrhea in 40–50% of patients—are predominantly mild to moderate and concentrate during the four to sixteen weeks of dose titration⁴³. Class-wide contraindications include personal or family history of medullary thyroid carcinoma or MEN2 syndrome. Acute pancreatitis is rare (approximately 0.2%), and cholelithiasis risk is elevated by rapid weight loss. The SELECT cardiovascular outcomes data for semaglutide decisively establish incretin-based agents as having mortality-modifying potential beyond anthropometric benefit⁵; the ongoing SURMOUNT-MMO trial is expected to provide analogous outcomes data for tirzepatide. Monitoring recommendations include baseline and three- to six-monthly complete blood count, metabolic panel, and HbA1c, with weight assessed every four weeks during titration¹¹. Achievement of at least 5% weight loss by twelve to sixteen weeks is the accepted threshold for predicting long-term response; non-responders should be reassessed for adherence, dose adequacy, or escalation to bariatric surgery.

Surgical and Procedural Treatments

Bariatric surgery represents the most potent and durable intervention currently available for severe obesity, producing sustained weight loss that far exceeds what is achievable through lifestyle modification or pharmacotherapy alone. Over three decades, the field has evolved from open procedures with substantial perioperative mortality to laparoscopic techniques with 30-day mortality rates comparable to common elective operations. Procedures once characterised purely by mechanical restriction or nutrient malabsorption are now understood to exert profound neuroendocrine and microbiome-mediated effects independent of caloric restriction.

Types of Bariatric Surgery

The principal bariatric procedures differ substantially in their anatomical modifications, mechanisms of action, magnitude of weight loss, and risk profiles. Roux-en-Y gastric bypass (RYGB) is widely considered the historical gold standard. It creates a small gastric pouch of 15–30 mL that is anastomosed to a Roux limb of jejunum, thereby bypassing the gastric fundus, antrum, duodenum, and proximal 100–150 cm of small bowel. This combined restrictive and malabsorptive anatomy produces 60–80% excess weight loss (EWL) at five years, together with dramatic remission of type 2 diabetes (T2D) and other metabolic comorbidities. The procedure's complexity and the permanence of its intestinal rerouting confer both superior metabolic efficacy and an obligate requirement for lifelong nutritional surveillance.

Sleeve gastrectomy (SG) has displaced RYGB as the most commonly performed bariatric procedure worldwide, now accounting for approximately 60% of all operations. SG is a purely restrictive procedure that removes approximately 80% of the stomach along the greater curvature, leaving a tubular gastric remnant of approximately 60–150 mL. The absence of an intestinal anastomosis simplifies the operation, reduces early complication rates, and preserves duodenal continuity, facilitating endoscopic access to the biliary tree post-operatively. Typical EWL is 50–70% at five years. Although SG lacks the intestinal bypass component of RYGB, post-operative reductions in ghrelin — a primary driver of hunger — and marked increases in GLP-1 and peptide YY (PYY) contribute substantially to appetite suppression and metabolic improvement beyond what restriction alone would predict.

Laparoscopic adjustable gastric band (LAGB) was introduced as a minimally invasive, fully reversible option providing 40–55% EWL, but has been largely abandoned owing to inferior long-term outcomes and reoperation rates exceeding 50% at ten years due to band slippage and erosion.

Biliopancreatic diversion with duodenal switch (BPD-DS) offers the highest degree of weight loss among standard bariatric operations, with EWL typically exceeding 70–80%. The procedure combines a sleeve gastrectomy with diversion of biliopancreatic secretions to a short common alimentary channel, producing profound malabsorption of macronutrients. This efficacy comes at the cost of the greatest risk of protein malnutrition, fat-soluble vitamin deficiency, and long-term metabolic complications; BPD-DS is therefore reserved for patients with BMI above 50 kg/m² or those with severe, treatment-refractory metabolic disease. The one-anastomosis gastric bypass (OAGB, or mini-bypass) is a more recently codified alternative that creates a long gastric tube anastomosed to a jejunal loop approximately 200 cm from the ligament of Treitz. Emerging data

suggest that OAGB achieves weight loss and T2D remission rates comparable to RYGB with lower operative complexity, and its adoption is growing internationally.

Long-term Outcomes: The Swedish Obese Subjects Study

The most comprehensive long-term dataset on bariatric surgery derives from the Swedish Obese Subjects (SOS) study, a prospective controlled trial enrolling 4,047 severely obese patients.⁸ reported that the surgical cohort maintained 16.1% weight loss at ten years compared with 1.6% in conventionally treated controls, while cumulative T2D incidence was reduced by 86% at eight years.³² analysed mortality in 2,010 surgical patients and 2,037 matched controls over a mean 10.9-year follow-up, finding an adjusted hazard ratio for all-cause mortality of 0.71 (95% CI 0.54–0.92; $p = 0.01$) — a 29% relative reduction.³³ subsequently demonstrated that surgery reduced the composite of fatal and non-fatal cardiovascular events with HR 0.67 (95% CI 0.54–0.83; $p = 0.0002$), driven principally by reductions in fatal MI and fatal stroke, establishing that bariatric surgery materially alters the natural history of cardiovascular disease.

Bariatric Surgery vs Intensive Medical Therapy for T2D

Several randomised controlled trials have directly compared bariatric surgery with intensive medical therapy for patients with obesity and T2D, consistently demonstrating superiority of the surgical approach for glycaemic control. The STAMPEDE trial⁷, which enrolled 150 participants with T2D and BMI ranging from 27 to 43 kg/m², provided five-year outcomes of three randomly assigned treatment arms: RYGB, sleeve gastrectomy, and intensive medical therapy. At five years, RYGB produced a mean weight loss of 23.4% from baseline, sleeve gastrectomy 19.0%, and intensive medical therapy 5.4%. The primary endpoint of HbA1c below 6.0% without antidiabetic medications was achieved by 29% of the RYGB group and 23% of the sleeve gastrectomy group, compared with only 5% of the medically treated group. Importantly, this superiority of surgical T2D remission persisted despite modest weight regain in the surgical arms between years two and five, indicating that metabolic benefits are not entirely explained by weight loss.

⁵² reported a complementary Italian RCT of 60 patients with long-standing T2D (mean duration approximately nine years) randomised to RYGB, BPD, or conventional medical therapy. At two years, complete T2D remission — defined as HbA1c below 6.5% with discontinuation of all antidiabetic medications — was achieved in 75% of RYGB patients and 95% of BPD patients, compared with 0% in the conventionally treated group. The rapidity of remission, often occurring within days to weeks of surgery and before substantial weight loss, pointed unambiguously toward weight-independent mechanisms. The Utah Gastric Bypass cohort study³⁹, which compared 9,949 patients who underwent gastric bypass surgery with 9,628 matched obese controls over a mean follow-up of 7.1 years, demonstrated a 40% reduction in all-cause mortality (18.7 vs 28.2 deaths per 10,000 person-years). The specificity of this survival benefit was especially pronounced for cause-specific mortality: T2D-related deaths were reduced by 92%, and cardiovascular deaths by 59%.

Mechanisms Beyond Restriction and Malabsorption

The neuroendocrine consequences of bariatric surgery extend far beyond simple caloric restriction or nutrient malabsorption. Following both RYGB and sleeve gastrectomy, dramatic post-prandial increases in GLP-1 and PYY are detectable within days of surgery, well before clinically

significant weight loss has occurred⁴⁴. These hormonal changes reduce appetite, improve insulin secretion and sensitivity, and contribute directly to T2D remission independently of weight-mediated effects. Emerging evidence also implicates the bile acid–FXR–TGR5 signalling axis, with post-surgical changes in enterohepatic bile acid circulation driving beneficial effects on hepatic glucose metabolism and energy expenditure. Fibroblast growth factors FGF-19 and FGF-21 are likewise altered by intestinal rerouting, further contributing to improved insulin sensitivity and lipid metabolism.

The role of the gut microbiome in mediating post-surgical metabolic benefits has been demonstrated elegantly in germ-free mouse experiments, in which animals colonised with intestinal microbiota from post-RYGB donors gained significantly less weight than those receiving pre-operative microbiota, despite consuming comparable diets. At the neural level, bariatric surgery remodels food reward circuitry, reducing behavioural preference for high-fat and high-sugar foods in a manner consistent with changes observed in mesolimbic dopamine pathways. The landmark meta-analysis of 22,094 patients by³¹ documented the breadth of metabolic improvement achievable through bariatric surgery: T2D resolved in 76.8% of patients, dyslipidaemia improved in 70%, hypertension resolved in 61.7%, and obstructive sleep apnoea resolved in 85.7% — findings that have been replicated and extended in subsequent prospective cohorts.

Patient Selection and Surgical Criteria

Patient selection criteria for bariatric surgery were first formalised in the 1991 NIH Consensus Statement, which established BMI thresholds of 40 kg/m² or 35 kg/m² with significant comorbidities including T2D, hypertension, obstructive sleep apnoea, or non-alcoholic steatohepatitis, combined with failure of at least six months of supervised weight management and absence of specific contraindications. These criteria have remained the operational standard in most health systems, though they are increasingly recognised as overly restrictive. Updated ASMBS/IFSO guidance issued in 2022 acknowledges that metabolic surgery should be considered for patients with T2D and BMI as low as 30–35 kg/m², and for patients of Asian ethnicity with BMI above 27.5 kg/m² in the presence of obesity-related comorbidities, reflecting the higher metabolic risk at lower absolute BMI in this population.

Perioperative safety has improved dramatically with the transition to laparoscopic techniques and the concentration of bariatric procedures in high-volume centres. The LABS-1 study⁵³, which prospectively enrolled 4,610 patients across ten US clinical centres, reported composite 30-day adverse event rates of 4.3% for RYGB, 2.1% for sleeve gastrectomy, and 1.0% for adjustable gastric banding, with an overall 30-day mortality of 0.3% — comparable to laparoscopic cholecystectomy. These outcomes exhibited a clear volume–outcome relationship, with higher-volume centres achieving significantly lower complication rates. Long-term nutritional management remains a non-negotiable component of post-surgical care;³⁴ guidelines specify lifelong supplementation with vitamins B12, D, iron, calcium, and folate, with monitoring intervals that intensify whenever weight regain or clinical symptoms suggest nutritional compromise.

Endoscopic and Minimally Invasive Procedures

Endoscopic bariatric therapies occupy the treatment space between pharmacotherapy and surgery. Endoscopic sleeve gastropasty (ESG) employs suture-based plication along the gastric greater curvature without incisions;⁵⁴ demonstrated that ESG achieves approximately 15% total body weight loss at one year with a favourable safety profile, and the procedure is used both as a definitive treatment in surgery-averse patients and as a bridge to reduce operative risk. Intragastic balloon systems produce 10–15% weight loss over six months before endoscopic removal and serve as a short-term, reversible adjunct. Transpyloric shuttle devices and endoluminal bypass sleeves remain investigational. The collective advantage of endoscopic approaches is reversibility and a lower procedural risk profile; their limitations are reduced durability, operator skill dependence, and a comparatively restricted long-term evidence base.

Discussion

Synthesis of Key Advances, 2016–2024

The period from 2016 to 2024 has witnessed four paradigm shifts that fundamentally altered the conceptual and therapeutic landscape of obesity medicine, representing a discontinuity from prior decades rather than incremental refinement.

The first and most consequential is the reconceptualisation of obesity from a condition attributable to willpower failure to a recognised chronic neurobiological disease driven by dysregulation of the hypothalamic–gut–adipose axis^{14,18}. Altered hypothalamic satiety circuits, reduced dopaminergic tone in reward pathways, and impaired leptin and insulin signalling persist even after weight loss, explaining the remarkable difficulty of long-term weight maintenance by volitional effort alone and providing the scientific foundation for pharmacological and surgical interventions that modulate appetite biology directly.

The second paradigm shift is the emergence of GLP-1 receptor agonists and dual GLP-1/GIP agonists as transformative pharmacotherapy. STEP 1 demonstrated that semaglutide 2.4 mg once weekly produced a mean 14.9% weight reduction over 68 weeks in patients with obesity without diabetes³, while SURMOUNT-1 showed tirzepatide 15 mg achieving 20.9% weight loss at 72 weeks⁴ — results that, for the first time, approached the 20–25% weight loss typical of sleeve gastrectomy and redefined expectations for pharmacological intervention.

The third paradigm shift is the demonstration that cardiovascular benefit from GLP-1 RA therapy is independent of glucose lowering. The SELECT trial⁵ enrolled 17,604 patients with pre-existing cardiovascular disease, obesity, and no diabetes; semaglutide 2.4 mg reduced the composite of cardiovascular death, non-fatal MI, and non-fatal stroke by 20% (HR 0.80, 95% CI 0.72–0.90; $p < 0.001$) over 34.2 months, establishing for the first time that treating obesity itself reduces hard cardiovascular endpoints.

The fourth paradigm shift is the recognition that metabolic dysfunction-associated steatohepatitis (MASH, formerly NASH) constitutes a first-line indication for pharmacotherapy targeting weight loss. Until 2024, no approved pharmacotherapy existed specifically for MASH. The SURMOUNT-NASH trial⁶, a phase 3 randomised controlled trial, demonstrated that tirzepatide 10 mg and 15

mg weekly over 52 weeks achieved MASH resolution without worsening of fibrosis in 44.4% and 62.4% of patients, respectively, compared with 9.7% in the placebo group, thereby opening a new therapeutic indication for an agent initially developed for glycaemic control.

Comparing Treatment Modalities

Across the full spectrum of interventions examined in this review, a consistent hierarchy of efficacy is apparent that tracks closely with the degree of biological mechanism engaged. Lifestyle modification alone — even when delivered with high intensity — achieves a mean weight loss of approximately 5–8% over one year, which, while sufficient to reduce T2D incidence in at-risk individuals, falls short of the 10–15% threshold generally associated with clinically meaningful improvement in cardiometabolic risk factors in patients with established obesity-related disease^{24,46}. The Look AHEAD trial's failure to demonstrate a reduction in cardiovascular events despite sustained 6–8% weight loss at eight years underscored that lifestyle intervention alone is insufficient to modify cardiovascular outcomes in patients with existing T2D and obesity.

Contemporary pharmacotherapy with GLP-1 RAs and the dual GLP-1/GIP agonist tirzepatide bridges a substantial part of the gap between lifestyle intervention and bariatric surgery. Mean weight losses of 15–21% are now routinely achieved in phase 3 trials, with a meaningful proportion of patients achieving 25% or greater weight loss on tirzepatide — outcomes that overlap substantially with the weight loss distribution following sleeve gastrectomy. Bariatric surgery remains the most effective single intervention for severe obesity, producing 50–80% EWL at five years depending on procedure, with the most potent metabolic effects observed with RYGB and BPD-DS^{7,31}. Crucially, surgical efficacy in achieving T2D remission appears to be partially independent of the degree of weight loss achieved, reflecting the unique neuroendocrine mechanisms operative after intestinal rerouting⁴⁴. The most clinically promising strategy emerging from recent literature is the combination of pharmacotherapy and surgical or endoscopic intervention — using GLP-1 RAs to optimise weight loss pre-operatively, reduce operative risk, manage weight regain post-operatively, or treat residual metabolic disease after bariatric procedures.

Special Populations

Several subpopulations warrant tailored consideration in the context of obesity pharmacotherapy and surgery. Among patients with coexisting T2D, the convergence of glycaemic control and weight loss within single agents is particularly valuable. Tirzepatide in SURMOUNT-2 achieved 15.7% weight loss and a mean HbA1c reduction of 2.4 percentage points in patients with T2D and obesity⁵⁰, while⁷ demonstrated that bariatric surgery achieves complete T2D remission in 23–29% of patients even at moderate BMI levels between 27 and 35 kg/m², reinforcing the metabolic indication for surgery below traditional weight thresholds. These complementary data support a framework in which pharmacotherapy and surgery are viewed as synergistic rather than competing strategies within a T2D treatment algorithm.

Patients with MASH represent an emerging priority population for whom no pharmacotherapy was previously approved. SURMOUNT-NASH⁶ establishes tirzepatide as the first agent to demonstrate both MASH resolution and anti-fibrotic efficacy in a phase 3 setting, addressing a condition

affecting an estimated 15 million people in the United States and constituting an increasing indication for liver transplantation.

In adolescents,³⁵ demonstrated significant BMI reduction with liraglutide 3.0 mg daily in a phase 3 RCT, while³⁶ reported that semaglutide 2.4 mg weekly achieved a 16.1% reduction in BMI versus 0.6% with placebo over 68 weeks — a treatment effect of remarkable magnitude. These data support a multidisciplinary approach integrating pharmacotherapy with behavioural intervention and family-centred care. For elderly patients, sarcopenic obesity creates a distinct challenge: weight loss that preferentially depletes lean mass may worsen physical function and increase fall risk, requiring resistance exercise, protein supplementation, and DEXA-based body composition monitoring alongside pharmacotherapy. The net benefit–risk balance of GLP-1 RAs in adults over 70 remains incompletely characterised and merits dedicated investigation.

The Weight Regain Challenge

Weight regain following cessation of pharmacotherapy is a defining challenge that reflects the persistence of neurobiological mechanisms defending an elevated body weight set point. The STEP 1 extension study²¹ demonstrated that participants who discontinued semaglutide regained approximately two-thirds of lost weight within one year, with near-complete reversal of cardiometabolic improvements. The SURMOUNT-4 trial⁴⁰ confirmed this with tirzepatide: continued therapy produced a further 5.5% weight loss, whereas patients switched to placebo regained 14% of body weight over 52 weeks. These data establish that pharmacological weight loss is contingent on continuous drug exposure — analogous to antihypertensive or statin therapy — and that obesity must be managed as a chronic condition requiring chronic treatment.

Precision Medicine in Obesity

Substantial heterogeneity in individual treatment responses motivates a precision medicine approach to obesity treatment selection. Genetic profiling is beginning to identify stratifying variants: MC4R mutations, the most common monogenic cause of obesity, may predict reduced response to agents acting downstream of the melanocortin pathway; GLP1R variants are associated with differential GLP-1 secretion and potentially variable semaglutide response; and common FTO locus variants are under investigation as risk-prediction biomarkers. Gut microbiome composition influences energy harvest, bile acid metabolism, and GLP-1 secretion in ways that likely modulate treatment response to both pharmacological and surgical interventions. Psychobehavioural phenotyping — distinguishing hedonic or emotional eating from hyperphagia driven by deficient satiety signalling — may also guide pharmacotherapy selection, as these subtypes may respond differentially to agents targeting reward circuitry versus those primarily suppressing meal-driven appetite.

Health Equity and Access

Transformative efficacy is accompanied by a stark access disparity. Tirzepatide and semaglutide 2.4 mg carry list prices of approximately \$1,000–1,500 per month without insurance in the United States, placing them beyond reach for populations bearing the highest obesity burden. As of 2024, Medicare did not cover GLP-1 RAs for obesity without T2D, and many Medicaid programmes similarly exclude these agents, effectively making access a function of employer-

sponsored insurance generosity. Globally, these agents are approved in a fraction of countries where obesity prevalence is highest. Generic availability for semaglutide and tirzepatide is likely a decade or more distant given current patent portfolios, meaning that access barriers will persist unless policy interventions mandate coverage or accelerate lower-cost alternatives.

Implications for Clinical Practice and Policy

The evidence synthesised here carries several direct implications for clinical practice. First, clinicians should treat obesity as a chronic relapsing disease requiring indefinite management rather than a time-limited weight-loss episode; the SURMOUNT-4 and STEP 1 extension data make clear that discontinuation predictably reverses benefit, so treatment duration should be framed with patients at initiation in the same terms used for antihypertensive or lipid-lowering therapy^{21,40}. Second, treatment intensity should be matched to complication burden rather than to BMI alone: the Edmonton staging and ABCD frameworks recommend that a patient with stage 2–3 disease at BMI 32 warrants more aggressive intervention than a metabolically healthy patient at BMI 36¹⁵. Third, in patients with obesity and established cardiovascular disease, semaglutide 2.4 mg should now be considered a cardiovascular risk-reduction therapy in its own right, not merely a weight intervention, given the 20% MACE reduction demonstrated in SELECT⁵. Fourth, structured lifestyle intervention should be maintained alongside pharmacotherapy rather than abandoned once a drug is initiated, since the SURMOUNT-3 and STEP 3 designs demonstrate additive benefit when the two are combined^{48,51}. Fifth, bariatric surgery referral should not be deferred indefinitely in patients with severe obesity and type 2 diabetes while successive pharmacological agents are trialled, given the durable remission and mortality benefit demonstrated over long-term follow-up^{7,32}.

At the policy level, the principal recommendation arising from this review is that coverage frameworks have not kept pace with the evidence. Reimbursement policies that exclude anti-obesity pharmacotherapy while covering treatments for the downstream complications of obesity are difficult to reconcile with cardiovascular outcome data, and payers and health systems should revisit exclusion criteria in light of the SELECT results. Health systems should also invest in multidisciplinary obesity care capacity — including dietetic, behavioural, and bariatric surgical services — since pharmacotherapy alone cannot deliver the combined outcomes demonstrated in trial settings that embedded structured lifestyle support.

Future Directions

The pharmacological pipeline beyond current GLP-1 RAs is exceptionally active. Retatrutide, a triple GLP-1/GIP/glucagon receptor agonist, produced a mean 24.2% body weight reduction at 48 weeks in a phase 2 trial³⁷, exceeding any previous pharmacological result and approaching RYGB outcomes; phase 3 trials are underway. Oral semaglutide is approved for T2D and under regulatory evaluation for the obesity indication, which would lower the barrier associated with subcutaneous injection. The amylin analogue cagrilintide in combination with semaglutide (CagriSema) has shown approximately 15% weight loss in phase 2, providing an additional mechanism for patients with partial GLP-1 RA responses. Setmelanotide, a melanocortin-4 receptor agonist, has established proof of concept in monogenic obesity syndromes. Gene therapy targeting FTO and MC4R dysregulation remains at preclinical stage. Continued refinement of ESG technique

and development of more durable endoluminal devices are expected to consolidate endoscopic procedures as a meaningful tier within the stepped care model.

Strengths and Limitations of This Review

Strengths of this review include a multi-database search strategy with CiteScore/SJR quality filtering, structured two-stage screening with pre-specified criteria, independent DOI validation via the doi.org resolution API, and integration of lifestyle, pharmacological, and surgical literatures enabling cross-modality comparisons absent from single-domain reviews. Limitations include network access restrictions that reduced Embase and Scopus coverage during portions of the search, potentially introducing partial database bias. The 2016–2024 search window, while capturing the most clinically relevant evidence, necessarily excludes foundational earlier trials; these were incorporated as background citations by design. Publication bias inherent to any systematic review — positive results being disproportionately published and indexed in quality-filtered journals — cannot be fully mitigated. The field is evolving rapidly, and phase 3 data for several pipeline agents will emerge after this review’s search cutoff.

Conclusions

Obesity has emerged from decades of therapeutic neglect to become one of the most actively researched conditions in contemporary medicine, and the period from 2016 to 2024 has delivered advances of a magnitude and pace rarely seen in any therapeutic area. The central message of this systematic review is that obesity is a complex, chronic, neurobiologically mediated disease for which effective treatments now exist across a broad spectrum of severity and patient circumstance. GLP-1 receptor agonists and the dual GLP-1/GIP agonist tirzepatide have achieved pharmacological weight reductions — 15–21% of body weight in pivotal trials — that rival those achieved by bariatric sleeve gastrectomy in selected patients, while the SELECT trial has established cardiovascular outcome benefit independent of glycaemic effect. Bariatric surgery remains the most potent and metabolically transformative modality available for severe obesity with comorbid T2D, and its superiority in achieving T2D remission over intensive medical therapy has been definitively confirmed in multiple randomised controlled trials. The challenge ahead is not primarily one of efficacy — the treatments now available are effective — but of access, durability, and personalisation. Ensuring that the most effective interventions reach the patients with the greatest need, managing the chronic nature of the disease with continuous long-term treatment strategies, and developing precision tools to match individuals to the most appropriate therapy represent the defining clinical and scientific priorities for the next decade of obesity medicine.

PRISMA 2020 Checklist

This systematic review was conducted and reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement. The following checklist documents compliance with each of the 27 items.

Section	Item	Checklist Item	Reported on
TITLE			
Title	1	Identify the report as a systematic review	Title page
ABSTRACT			
Abstract	2	Structured summary (background, objectives, methods, results, conclusions)	Abstract
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge	Section 1, para 1–3
Objectives	4	Provide an explicit statement of the objectives or questions the review addresses	Section 1, objectives list
METHODS			
Eligibility criteria	5	Specify inclusion and exclusion criteria for the review	Sections 2.4, 2.5
Information sources	6	Specify all databases, registers, websites, and other sources searched	Section 2.2
Search strategy	7	Present the full search strategies for all databases	Section 2.2
Selection process	8	Specify methods used to decide whether a study met inclusion criteria	Section 2.3
Data collection process	9	Specify methods used to collect data from reports	Section 2.7
Data items	10a	List and define all outcomes for which data were sought	Section 2.7
	10b	List and define all other variables for which data were sought	Section 2.7
Study risk of bias assessment	11	Specify methods used to assess risk of bias in included studies	Section 2.6
Effect measures	12	Specify for each outcome the effect measure (or measures) used to analyse the data	Section 2.7 (narrative synthesis)
Reporting of results and other materials		Identify all data sources used in the analysis	Section 2.7 (narrative synthesis) outputs are publicly available at https://github.com/ymeroom/robust-lit-review

Checklist adapted from: Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*. 2021;372:n71.

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